



INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH (IISER) BHOPAL

INTERNSHIP REPORT

PCB Layout Design for Cyclic Voltammetry Board

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Acknowledgment:

I would like to express my heartfelt gratitude to Dr. Santanu Talukder, Supervisor at Sixth Sense Lab, Indian Institute of Science Education and Research (IISER), Bhopal, for providing me with the opportunity to work on this exciting and enriching project. His expert guidance, constant encouragement, and valuable feedback played a vital role throughout the internship.

I am sincerely thankful to Prof. Debjit sir, for his patient explanations and for introducing me to the fundamentals of cyclic voltammetry and electrochemical systems. His mentorship helped me understand the theoretical concepts and translate them into practical designs.

I also extend my appreciation to my team members — Khushi, Aravind, Shreya, Onkar, and Gopika — for their collaboration and support during the project.

Lastly, I am grateful to the faculty and staff of Yeshwantrao Chavan College of Engineering, Nagpur, and the Department of Electrical Engineering and Computer Science (EECS) at IISER Bhopal for their academic support and for facilitating this internship opportunity.

This experience has significantly contributed to my technical knowledge and personal growth, and I feel fortunate to have been a part of this internship.

Abstract:

This report documents the work carried out as part of a research internship at the Sixth Sense Lab, IISER Bhopal, focusing on the design and development of a Printed Circuit Board (PCB) layout for cyclic voltammetry (CV) board.

Cyclic voltammetry is a widely used electrochemical technique for studying redox reactions and the electrochemical behavior of various substances. The goal of this project was to develop a compact, cost-effective, and efficient analog front-end capable of detecting low-level currents generated during electrochemical processes, particularly within the specific microampere range.

Introduction and Initial Briefing:

During the first meeting at the lab, Debjit sir (Ph.D. Scholar) explained the fundamentals of cyclic voltammetry (CV) and its applications in electrochemical sensing. Debjit sir (Ph.D. Scholar) further oriented us towards the goal of designing a compact, analog cyclic voltammetry system.

We were introduced to the working of electrochemical systems. The team was briefed about the requirements and performance expectations of each functional block of the cyclic voltammetry system.

This helped us understand the basic theory and real-world constraints in designing electrochemical measurement systems.

The CV board is used to apply controlled voltage sweeps to an electrochemical cell and measure the resulting current, enabling the study of redox reactions and electroactive species.

In electrochemical measurements, two main configurations are commonly used:

- **2-electrode system:** Consists of a working electrode and a **counter/reference electrode**. It is simpler but less accurate for precise measurements, as it combines reference and counter electrode functions, leading to potential control issues.
- **3-electrode system:** Consists of a working electrode, a reference electrode, and a counter electrode. This configuration provides more stable and accurate measurements by isolating the current path (counter electrode) from the potential measurement path (reference electrode).

PCB Design: RC Phase Shift Oscillator:

As a preliminary task, we were assigned to design a PCB for an RC Phase Shift Oscillator using KiCad. This exercise was intended to help us become familiar with the PCB design workflow, including schematic creation, component placement, routing, and layout generation. It served as an introductory step to understand the features and tools offered by KiCad before moving on to more complex circuit designs.

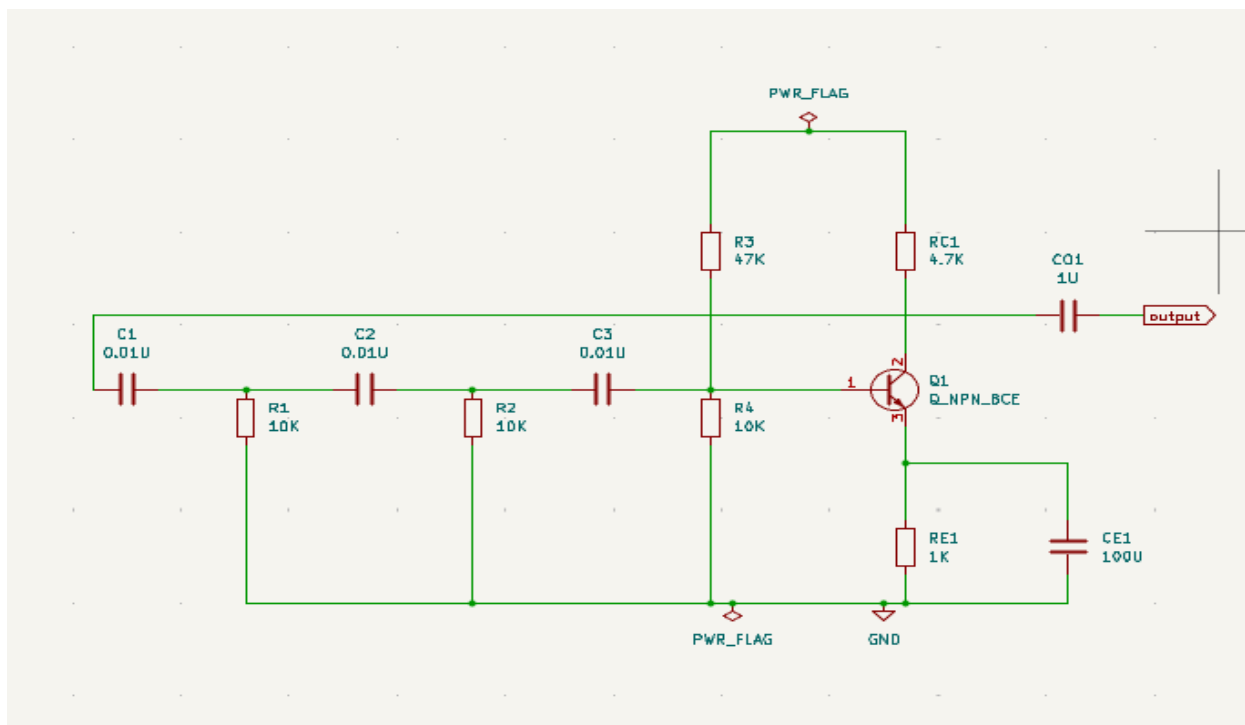


Figure 1: Schematic diagram of RC Phase shift oscillator.

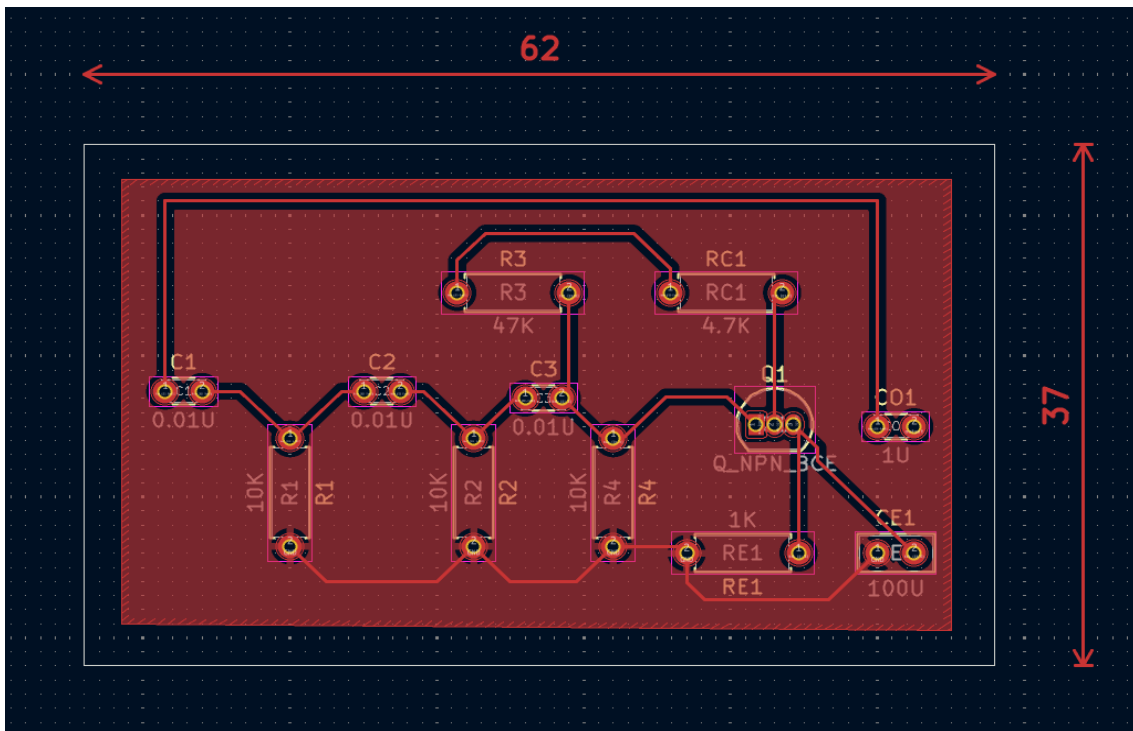


Figure 3: PCB Layout of RC phase shift Oscillator

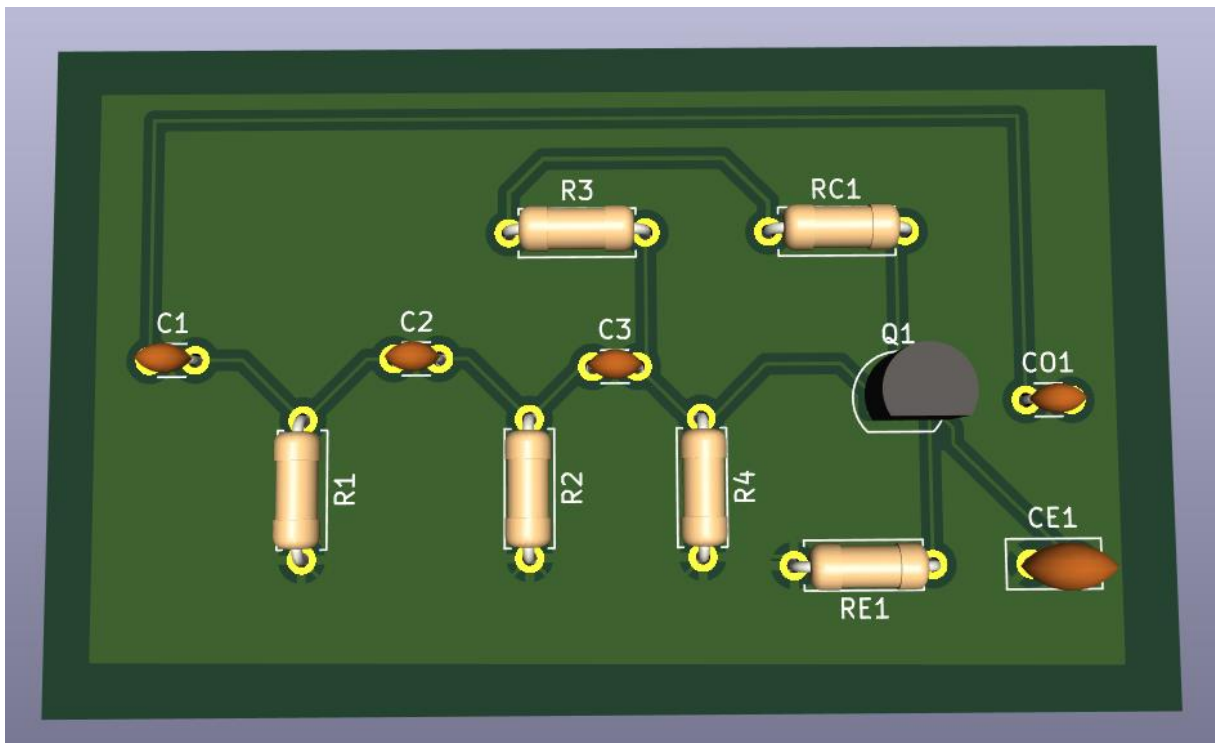


Figure 4: 3D view of PCB (RC phase shift oscillator)

Team Allocation and Current Sensing Circuit

After the initial orientation, our group was divided into two teams to work on different functional blocks of the cyclic voltammetry system:

- Current Sensing Circuit Team: Khushi, Aravind, Dhruv
- Pulse Generating Circuit Team: Shreya, Onkar, Gopika

Our team was assigned the task of designing a **current sensing circuit** using a Transimpedance Amplifier (TIA) configuration. The objective was to accurately detect low-level currents in the range of **1 μA to 200 μA** , which are typically generated during electrochemical reactions in cyclic voltammetry.

To support the design, we first studied the structure and operation of 2-electrode and 3-electrode systems commonly used in electrochemical measurements. The 2-electrode system, consisting of a working and a counter/reference electrode, is simpler but less accurate. The 3-electrode system, which includes a separate reference electrode, provides more precise voltage control and is ideal for accurate current measurements.

The TIA-based current sensing circuit was simulated using LTSpice to validate its performance. After successful simulation, the design was implemented in KiCad as a PCB layout, preparing it for integration into the complete cyclic voltammetry board.

Working Principle:

In cyclic voltammetry, the current response from the electrochemical cell is typically very small, usually in the microampere range. To accurately detect this small current, a Transimpedance Amplifier (TIA) configuration is used.

- In a TIA, the input current(I_{in}) is converted into a proportional voltage(V_{out}) using an operational amplifier (LM158) with a feedback resistor (R_f).
- The basic relationship is given by:

$$V_{out} = -I_{in} \times R_f$$

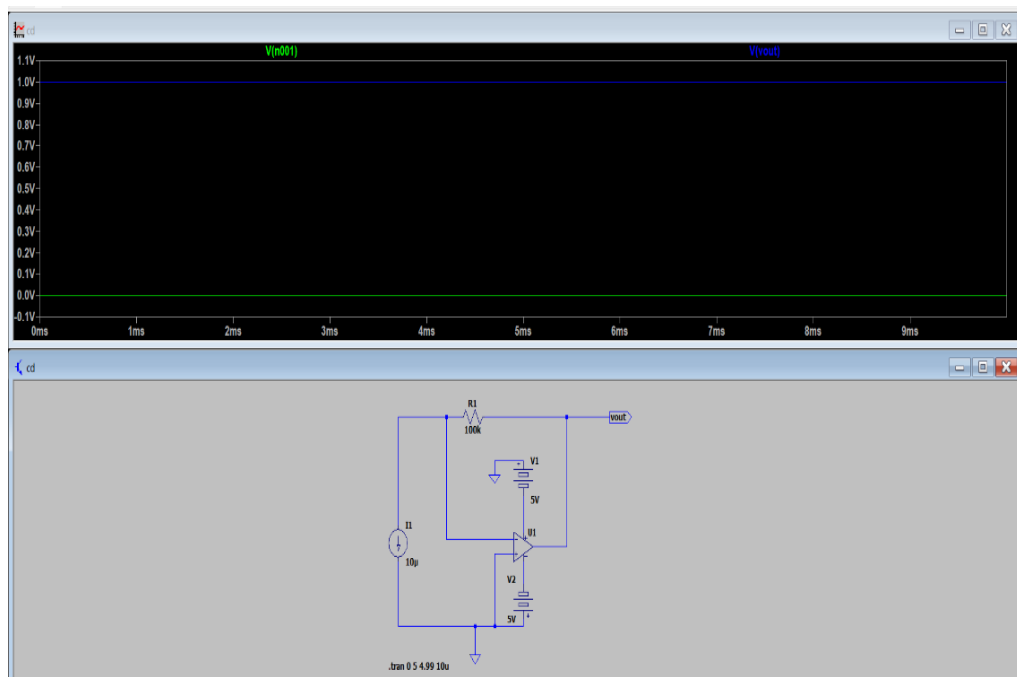


Figure 4: LTspice simulation of the Current Sensing Circuit

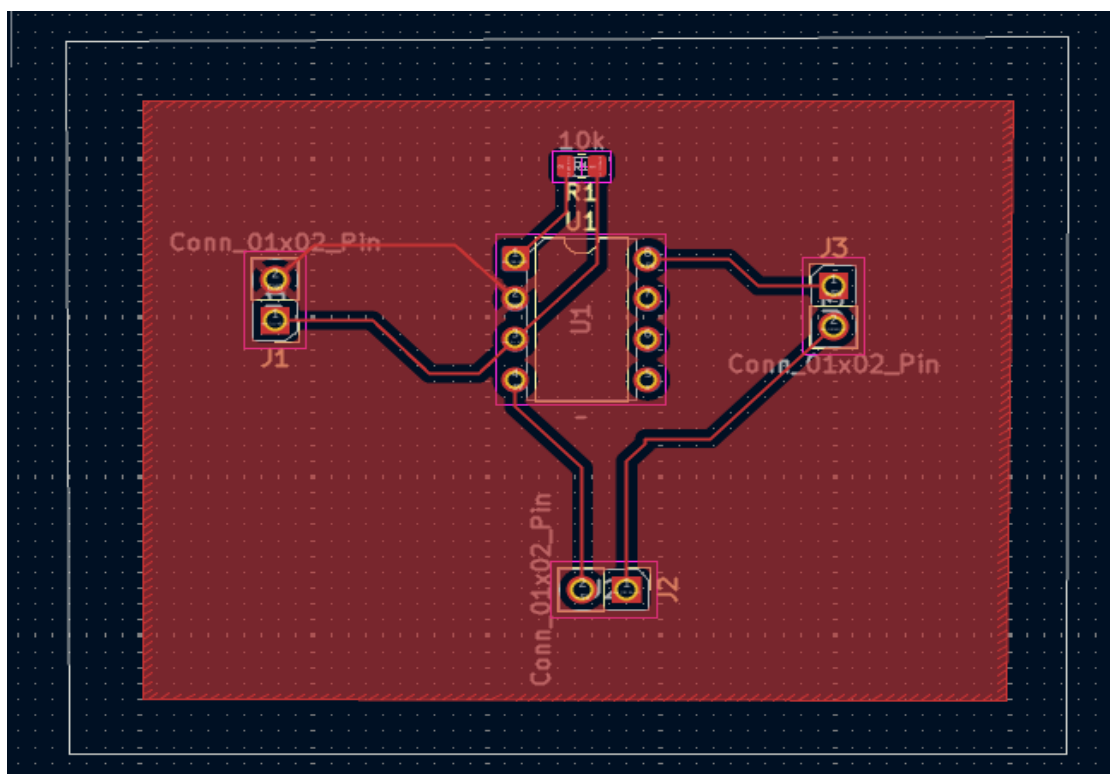


Figure 5: PCB Layout of Current Sensing Stage.

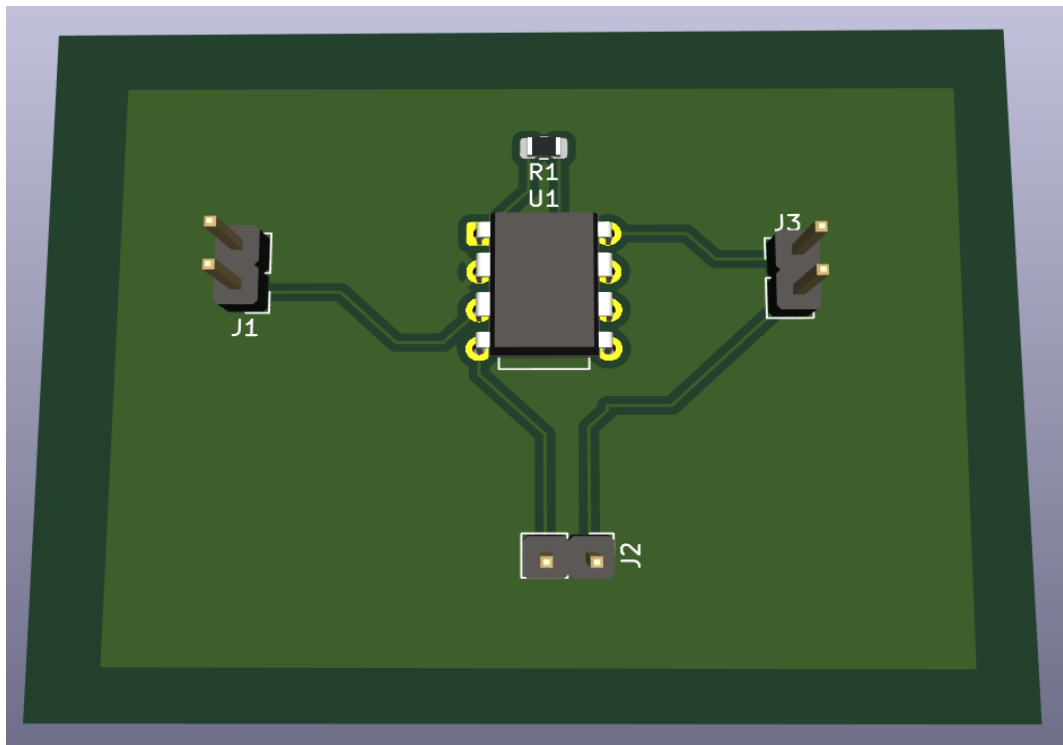


Figure 6: KiCad PCB design 3D view.

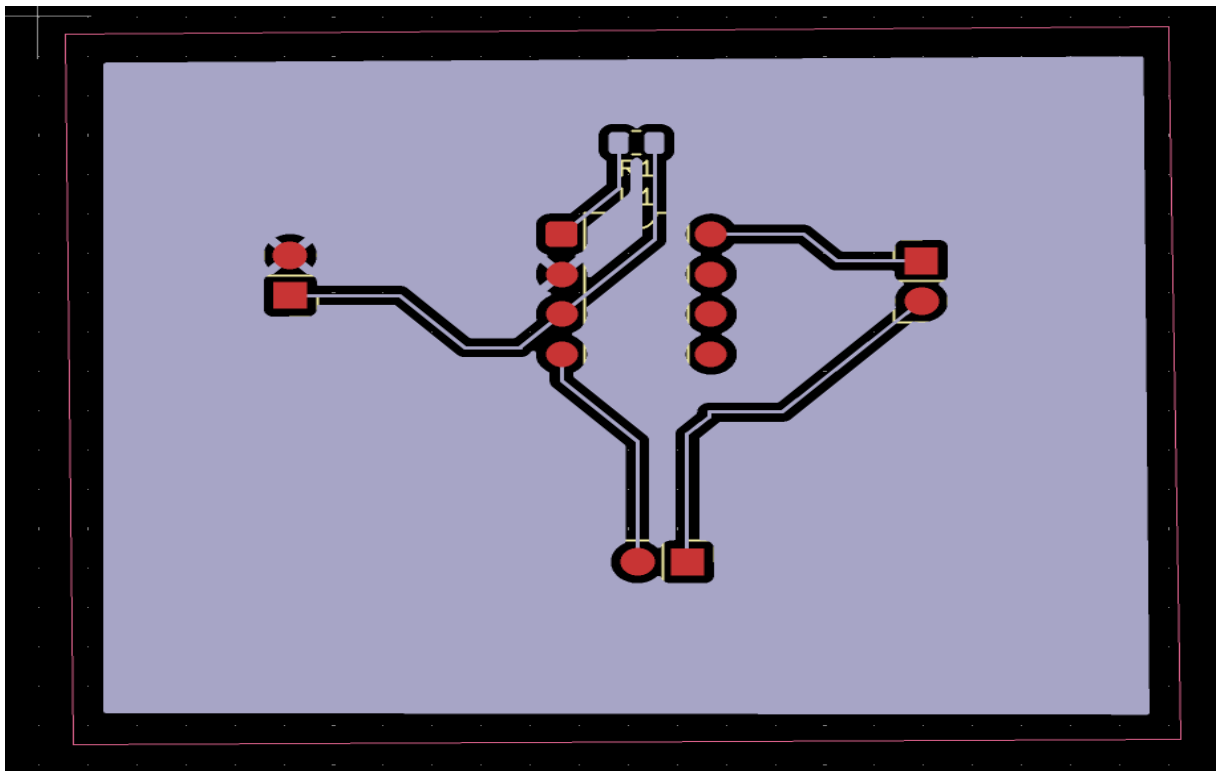


Figure 7: Gerber view

Results and Observations:

If $R_f=100\text{k}\Omega$ and input current(I_{in}) is in the range of $1\text{ }\mu\text{A}$ to $200\text{ }\mu\text{A}$ in upward direction, then:

- For $1\text{ }\mu\text{A}$: $V_{out} = -(1\text{ }\mu\text{A} \times 100\text{k}\Omega) = -100\text{mV}$
- For $200\text{ }\mu\text{A}$: $V_{out} = -(200\text{ }\mu\text{A} \times 100\text{k}\Omega) = -20\text{V}$
- When current is in downward Direction, Output voltage = (Input current $\times$ Feedback resistor)
- Example: If input current = $1\text{ }\mu\text{A}$ and feedback resistor = $100\text{k}\Omega$, then output voltage = $(1\text{ }\mu\text{A} \times 100\text{k}\Omega) = 100\text{mV}$

This voltage output can then be further processed, displayed, or digitized for analysis in the CV system.

Progress Review and Documentation

In the final meetings, Debjit sir reviewed our circuit designs and asked for justification of the op-amp specifications. We were also asked to submit:

- Technical Report
- Presentation slides (PPT)
- Updated schematics and layout snapshots.

My Key Learnings:

- Learned complete workflow of KiCad: schematic design, PCB layout, and 3D view
- Got familiar with LTSpice for analog circuit simulation and waveform analysis
- Understood the concept of 2-electrode and 3-electrode systems used in electrochemical analysis
- Designed circuits like the RC Phase Shift Oscillator and Current Detection (TIA).
- Developed a low-current sensing circuit ($1\text{ }\mu\text{A}$ to $200\text{ }\mu\text{A}$) using a Transimpedance Amplifier.